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F78AB
Actuarial and Financial Mathematics

Chapter 11: Fixed Interest and Index-Linked Securities

LEADERS IN IDEAS AND SOLUTIONS

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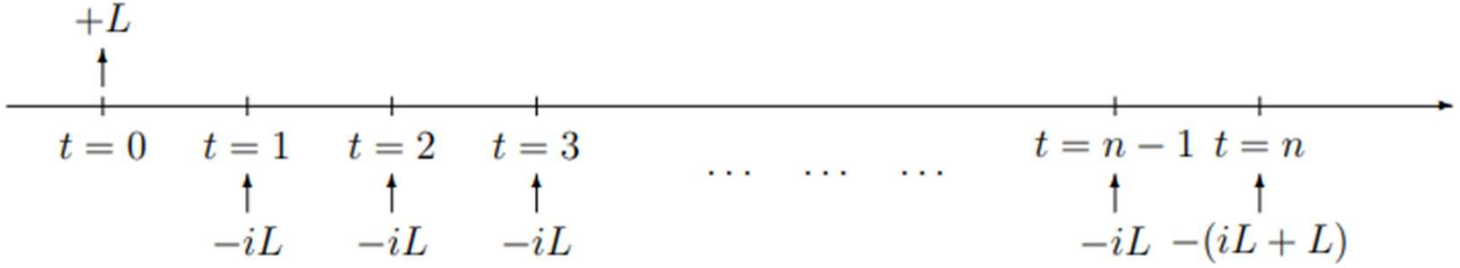
Introduction and Terminology

- A government or corporation can raise money in the capital markets by issuing **fixed interest securities** (FIS), also called **bonds**.
- Investors will lend money to the **issuer** (**borrower**) and in return will receive fixed interest payments at fixed dates plus repayment of the loan at the end of the term.

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Introduction and Terminology

- Time



- The loan is usually very large, millions or even billions of currency units.
- The loan will be split into smaller units that can be traded on an exchange.

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Introduction and Terminology

- The **nominal amount** (or face value) of a bond is the amount of loan it represents. It is **not** the price of the bond.
- For example, a company may raise £100 million by issuing 1 million bonds each of £100 nominal. The market will determine the price.
- Even at **issue** (the start date for the bond), the price could be different from the nominal amount.

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Introduction and Terminology

- The **coupon** is the interest that is paid on the bond and is normally expressed as a percentage per year of the nominal amount.
- Coupons are **commonly** paid twice a year (half-yearly) in arrears.
- For example, a bond of nominal value £100 that pays coupons at 6% per annum payable half yearly means that for each £100 nominal a coupon of £3 is payable every half-year.

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Introduction and Terminology

- The loan is repaid or **redeemed** at the end of the term.
- The date at which the loan is redeemed is called the **redemption date** (or **maturity date**).
- The **redemption rate** is the amount per £100 nominal that is repaid, and it is expressed as a percentage.

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Introduction and Terminology

- A bond may be redeemed at **par**, **discount** or **premium**.
- A bond redeemed at 100% is **redeemed at par**.
- A bond is **redeemed at a discount** if it is redeemed at less than 100%.
- A bond is **redeemed at premium** if it is redeemed at more than 100%.

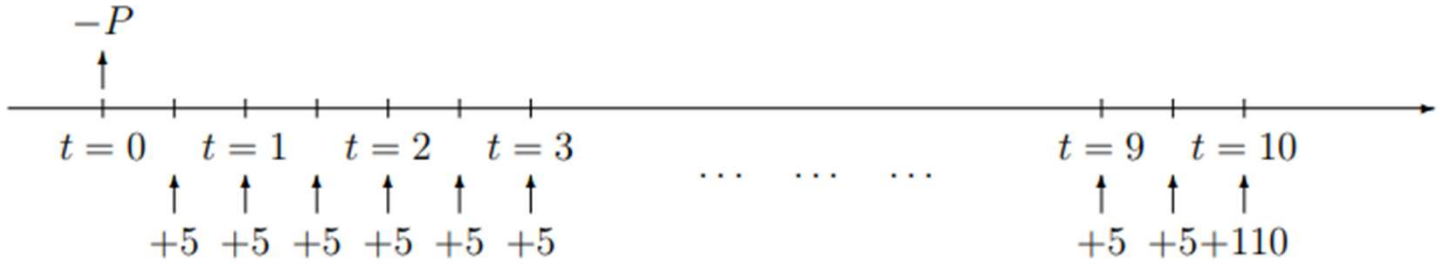
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Introduction and Terminology

E.g

A company issues a 10-year bond to be redeemed at 105% (i.e. 5% above par OR at a premium of 5%), with coupons of 10% per annum payable half-yearly. The nominal amount of each bond is £100.

Draw



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Bonds are issued under many different names.

Issued by	Called
UK Government	Gilts
US Government	Treasuries or T-bonds
Corporations	Debentures
Public sector corporations	Loan stocks
High risk bonds issued by debt-financed companies	Junk bonds

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Bond Prices

- Bonds do not have to be held to maturity
- They can be bought and sold in the ‘secondary market’
- At any time, including the issue date, investors will decide what price to pay for a bond.
- This is the basis of buying and selling bonds in the exchange.
- An important criterion for determining the price investors are prepared to pay is the yield they can obtain on the bond, compared to the yield available on other financial assets.

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Bond Prices

E.g. A company issues a 10-year bond to be redeemed at 105%, with coupons of 10% payable half-yearly. An investor wishes to buy the bond on its issue date to give a yield of 11% per annum effective. What price should the investor pay?

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Comments

- The nominal amount of the loan is just a theoretical figure on which the coupon and redemption rates are based (it is usually £100).
- The amount of capital the borrower can actually raise on the issue date is the price that investors are willing to pay for the future income stream.

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Comments

- As well as the yield on alternative investments, investors will consider the **credit risk of the issuer**, that is, the risk that the bond issuer might default on interest or capital payments.
- There is an inverse relationship between yields and bond prices (i.e. higher yield implies lower price). – think about the mathematics behind this.

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Comments

- Once a bond is issued, it can be traded on an exchange at any time until it is redeemed.
- The price will depend on
  - the remaining term to redemption; and
  - market conditions such as the yields obtainable on other investments

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Bond Prices

E.g. cont..

Just after the 10th coupon has been paid, the investor sells the bond. At that time, the market for comparable 5 year bonds is indicating a yield of 7% per annum effective.

a) What price does the investor obtain for the bond?

b) What yield has the investor achieved on the completed transaction?

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9.3 Bond Yields

There are two important yields on bond transactions.

1. The **redemption yield** is the yield obtained by buying a bond and holding it to redemption. If all taxes are ignored this is known as the **Gross Redemption Yield** (GRY). In the last example, the investor who bought the bond on the issue date would have obtained a GRY of 11% per annum had she held it to redemption, while the investor who bought the bond from her after five years would obtain a GRY of 7% per annum if they held it to redemption. *You should check these calculations!*

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Bond Yields cont…

2. The **realised yield** is the yield the investor obtains when they sell the bond before redemption.

The realised yield in the above example is 13.6% as we calculated above.

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Bond Yields

The main calculations involving bonds are:

- Given a required yield, or current yield in the financial markets; what is the price?
- Given the buying price, what is the redemption yield?
- Given both buying and selling prices, what is the yield on the completed transaction?

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Income Tax

- In the UK (and most other countries), **coupon payments** are regarded as **income** by the tax authorities.
- If the holder of a bond is liable to pay income tax, their net coupon income is reduced by the amount of income tax payable.
- Income tax is not levied on sale proceeds or redemption payments.
- Tax will affect both yields and prices.

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Income Tax

E.g.

An investor who pays income tax at 20% on all their income buys, at issue, a 25-year bond to be redeemed at par bearing semi-annual (twice per year) coupons of 8% per annum.

What price will the investor pay per £100 nominal to obtain a redemption yield, after income tax, of 9% per annum effective?

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Capital Gains Tax (CGT)

- CGT is payable on any gain made on the sale or redemption of a bond.
- The gain is defined as the difference between the price received on sale or redemption and the price originally paid.
- There is no relief from CGT on losses.

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Capital Gains Tax (CGT)

E.g.

On 1 January 2012 an investor buys two bonds:

- Bond A: Price = £101
- Bond B: Price = £99.

After one year, the investor sells both bonds for £100 each. How much capital gains tax does he pay, if the rate of CGT is 30%?

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Capital Gains Tax (CGT)

- As there is no rebate of CGT on losses, the equation used to find the price of a bond varies depending on whether CGT is payable.

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Capital Gains Tax (CGT)

Some notation:

- n = the outstanding term of a bond *(in units of coupon payment time)*
- g = the amount of each coupon payment per £100 nominal, before income tax
- P = the price paid per £100 nominal
- R = the sale or redemption amount per £100 nominal, before any CGT
- t_1 = the rate of income tax
- t_2 = the rate of capital gains tax
- $\acute{\imath}$ = the effective yield per *(coupon)* time unit obtained after all taxes.

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Capital Gains Tax (CGT)

- Case 1: R ≤ P so no CGT is paid

$$P = (1 - t_1)\, g a_{\overline{n}|}^{\,i} + R v^{n@i}$$

$$P(1+i)^n = (1 - t_1)\, g s_{\overline{n}|}^{\,i} + R$$

- Case 2: R > P so CGT is paid

$$P = (1 - t_1)\, g a_{\overline{n}|}^{\,i} + \left[R - t_2 (R - P) \right] v^{n@i}$$

$$P(1+i)^n = (1 - t_1)\, g s_{\overline{n}|}^{\,i} + \left[R - t_2 (R - P) \right]$$

Capital Gains Tax (CGT)

E.g.

An investor liable to pay CGT at rate 30% buys a 15 year bond with an annual coupon of 9% payable annually, to be redeemed at par. The price paid is £95 per £100 nominal.

What net redemption yield does the investor obtain?

Solution:

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EG Continued

An investor liable to pay CGT at rate 30% buys a 15 year bond with an annual coupon of 9% payable annually at £100 nominal, to be redeemed at par.

What price should the investor pay to obtain a net redemption yield of

a) 8% per annum effective?

b) 10% per annum effective?

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Capital Gains Tax (CGT)

- Capital Gains Tax is payable if and only if: $i > \frac{(1-t_1)g}{R}$

– If the yield was $(1 - t_1)g$ per time unit, the price would be P=100 and no CGT would be payable.

– If the yield was less than $(1 - t_1)g$, the price would rise, so P>100 and no CGT would be payable.

– If the yield was greater than $(1\text{-}t_1)g$, the price would fall, so P<100 and CGT would be payable.

Note: CGT would be less than the gain 100-P, so that CGT itself would not negate the gain.

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Capital Gains Tax (CGT)

E.g.

Consider a bond with term 10 years, coupon 10% per annum payable half-yearly, to be redeemed at 120%. Find the price, per £100 nominal, to give a net redemption yield of 7% per annum effective, to an investor subject to CGT at rate 25%, and:

a) Subject to income tax at 30%.

b) Not subject to income tax.

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Index-linked Gilts

- A special type of bond – where the coupon payments and redemption value are not fixed.
- Instead they are based on inflation, as measured by RPI
- The RPI figure used to determine the payment includes a time lag (ie is based on RPI months before the payment is made)
- Time lag is necessary because RPI figure for the month of payment is not available until after the month of payment

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Calculating the coupon and redemption payments

$$\text{Money value} = \text{Coupon/redemption payment} \times \frac{\text{Lagged RPI}}{\text{Base RPI}}$$

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What is the time lag?

- Index-linked gilts issued on or after 2005, the base RPI figures is that **3 months** before the issue date.
- The money values for coupon and redemption are normally rounded down to **six** decimal places.

- For I-L gilts issued earlier, the time lag is 8 month, and money values are normally rounded down for **four** decimal places

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Example 1 (from notes)

Example 11.7.2 The $4\frac{1}{8}\%$ Index-linked Treasury Gilt 2030 had an issue date of 12 June 1992 and uses an 8-month time lag. Its base RPI figure is 135.1, the RPI for October 1991.

It pays coupons on 22 January and 22 July each year. The latest coupon payments are shown in Table 11.1.

Coupon payment date	22 Jan 2009	22 Jul 2009	22 Jan 2010	22 Jul 2010	22 Jan 2011
Lagged RPI date	May 2008	Nov 2008	May 2009	Nov 2009	May 2010
Lagged RPI	215.1	216.0	212.8	216.6	223.6
Coupon payment (£ per £100 nominal)	3.2838	3.2975	3.2487	3.3067	3.4135

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Example 1 continued

- How are those recent coupon payments calculated?
- Let's look at the 22 July 2011 payment:

$$\frac{4\frac{1}{8}}{2}\times\frac{\text{RPI(Jul 2011 }-\{8\text{ months}\})}{\text{Base RPI figure}}=2.0625\times\frac{\text{RPI(Nov 2010)}}{\text{RPI(Oct 1991)}}=2.0625\times\frac{226.8}{135.1}=3.4624352.$$

- Rounded *down* to 4 d.p. the coupon payment is £3.4624 per £100 nominal
- Check the other coupon payments to ensure you understand where the figures come from

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Example 2

- Sometimes we will want to use the future rates of inflation (eg to determine the yield on the bond if we hold it until maturity)
- In this case we will need to estimate future RPI based on a given inflation rate.

Suppose today is 28 November 2013, and the last published RPI on Oct 2013 is 251.9. Assuming that RPI increases in the future a a constant rate of 2.5%p.a, calculate the projected RPI for November 2013, December 2013, January 2014, February 2014

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Example 3 (from notes)

Example 11.7.4 Suppose today is 22 January 2013 and the last published RPI figure is RPI(Dec 2012)= 246.8.

Today's market price of £100 nominal of the $4\frac{1}{8}\%$ Index-linked Treasury Stock 2030 is £323.57, just after the payment of the January 2013 coupon. The $4\frac{1}{8}\%$ Index-linked Treasury Stock 2030 was issued in June 1992 and has an 8-month time lag. Its Base RPI figure is 135.1, the October 1991 RPI.

Coupon dates: 22 January, 22 July. Redemption date: 22 July 2030.

Calculate the **money** yield-to-redemption assuming that RPI will increase in the future at a constant effective rate of 2.5% per annum.

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Example 3 cont..

Solution To calculate the money yield-to-redemption we need to:

1. Calculate the future money coupon and redemption payments by projecting the last published RPI figure at the assumed inflation rate, and

2. Solve the equation-of-value for the annual money yield i.

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Real yields on I-L Bonds

- As well as calculating the money yield to redemption on these I-L bonds, we may wish to calculate the real yield. (See Example 11.7.5 from Notes, and Tutorial 8)
- We do this in the same way as we calculated real yield in Chapter 5 – by determining the real value of each cash flow.
- So the steps will be:
  1. Given the bond-specific information, determine the money coupon and redemption payments, adjusted for time-lagged inflation as required
  2. Determine the real value of those money payments, adjusted for actual inflation over the relevant period

Do not conflate/confuse the adjustments in these two steps, they are different due to the time lag as specified in the bond terms.

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Real yields on I-L Bonds

- This difference means that the inflation protection from an I-L Bond is not perfect.
- It would only be perfect protection if there was no lag in the inflation adjustment for the bond.
- If there was no lag:
  - the real yield would be independent of future inflation, and depend only on market price
  - the real yield would be known as soon as the investor buys the bond
- However, this lag is necessary for practical reasons, which means that:
  - The real yield depends on future inflation as well as market price; and
  - The investor is exposed to some inflation risk (the longer the lag, the greater the exposure to risk)

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Break-even inflation rate

- I-L bonds give an indication of the market consensus on future inflation
- Consider the difference between two bonds:
  1) FI Bond with a coupon of £*C p.a.* payable *m* times a year and redeemed at *R%* in *N* years
  2) I-L Bond with annual coupon of £*C p.a.* payable *m* times a year and redeemed at *R%* in *N* years

The Break-even inflation rate = market consensus on future inflation (ignoring time lags etc)

= Money yield on the FI Bond – Real yield on the I-L Bond

At these yields the investor is indifferent between hodling the FI Bond or the IL Bond

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Attendance matters!

